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Detailed Performance Comparison of a Dump and Short Faired Combustor Diffuser System

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A rectangular model simulating four sectors of a combustion chamber was used to compare the performance of a standard dump diffuser, of overall length 180 mm, with that of a faired design 25.5 mm shorter. The performance of each system was assessed in terms of total pressure loss and static pressure recovery between prediffuser inlet and the annuli surrounding the flame tube. Since the program objective was to test design concepts only, no allowance was made for the presence of burner feed arms or flame tube support pins. In addition, tests were performed with relatively low levels of inlet turbulence and no wake mixing effects from upstream compressor blades. Relative to the dump design, the mass weighted total pressure loss to the outer and inner annuli was reduced by 30 and 40 percent, respectively, for the faired diffuser. Measurements around the flame tube head were used to identify regions of high loss within each system and account for the differences in performance. Within a dump diffuser the flow separates at prediffuser exit resulting in a free surface diffusion around the flame tube head and a recirculating flow in the dump cavity. This source of loss is eliminated in the faired system where the flow remains attached to the casings. Furthermore, the faired system exhibited similar velocity magnitudes and gradients around the combustor head despite its shorter length.

Introduction

One of the main objectives in aircraft engine design is to minimize weight. In the context of the combustor, this has led to the requirement for minimum length consistent with an acceptable total pressure loss and outlet temperature distribution. Since reducing the combustor burning length is prohibited by the need to meet pollution regulations, attention has therefore been focused on the diffuser system.

The primary function of a gas turbine combustor diffuser system is to distribute air issuing from the compressor around the flame tube. In current advanced core engines this objective is usually met with dump diffuser systems, which incorporate a prediffuser upstream of a recirculating dump region. Although such systems have relatively high losses, they are favored because of their inherent flow stability and insensitivity to manufacturing tolerances and thermal expansions. Such systems were initially investigated by Fishenden and Stevens [1] who monitored overall system performance while varying the prediffuser area ratio, dump gap, and mass-flow split to the inner and outer annuli. All tests were conducted with a flame tube that had a radial depth, expressed in terms of prediffuser inlet height (W/h_o), of 3.5. It was concluded that the principal determinants of total pressure loss in such systems are the amount of diffusion being attempted and the radius of curvature undertaken by the flow as it passes around the flame tube. This latter effect is a function of the size and shape of the flame tube and the dump gap. The more recent work of Young et al. [2], Stevens et al. [3], and Srinivasan et al. [4] illustrates the trend for flame tubes of increasing depth with values (W/h_o) in excess of 4.0. This implies a significant increase in the amount of turning and curvature that must be undertaken by the flow in the vicinity of the combustor head. Hence, although these modern systems have been optimized to produce an acceptable pressure loss in the shortest length possible, further reductions in length or increases in flame tube depth will result in relatively large increases in pressure loss. One possible method of maintaining system performance while achieving significant length reductions is therefore to define an alternative diffuser geometry.

A facility has been built at Loughborough (Fig. 1) that allows diffusers of different geometries to be tested. All models are four burner sectors wide and of rectangular cross section. Thus, it should be noted that diffusion of the flow in the circumferential direction present within annular systems is not simulated here. However, it is thought the relative performance of different diffusers can be accurately assessed while the rectangular geometry allows a large number of designs to be tested relatively easily. This paper outlines results relating to a standard dump diffuser (Fig. 2*a*), which is typical of current engine design practice. Its performance is compared with a faired design, which is 25.5 mm shorter (Fig. 2*b*), this representing

Contributed by the International Gas Turbine Institute and presented at the 38th International Gas Turbine and Aeroengine Congress and Exposition, Cincinnati, Ohio, May 24–27, 1993. Manuscript received at ASME Headquarters March 17, 1993. Paper No. 93-GT-331. Associate Technical Editor: H. Lukas.

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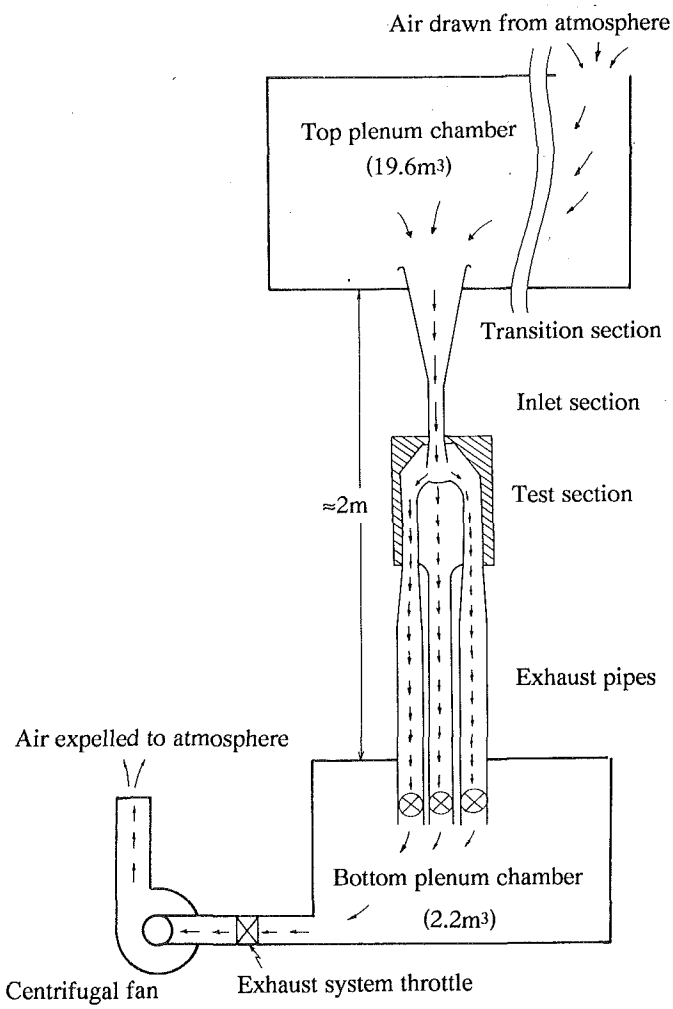


Fig. 1 Test facility—airflow paths

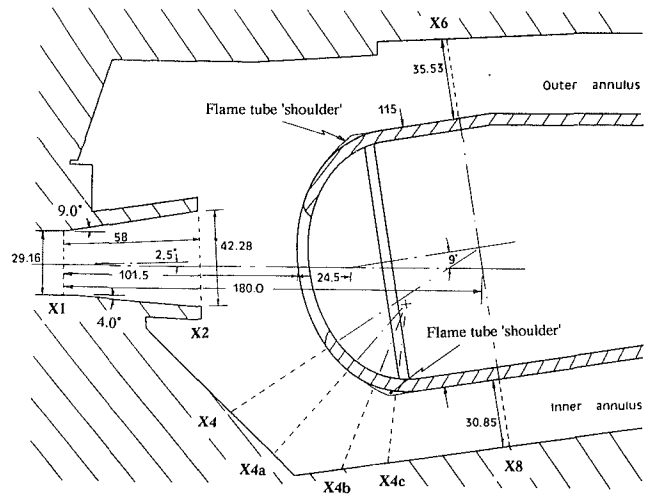
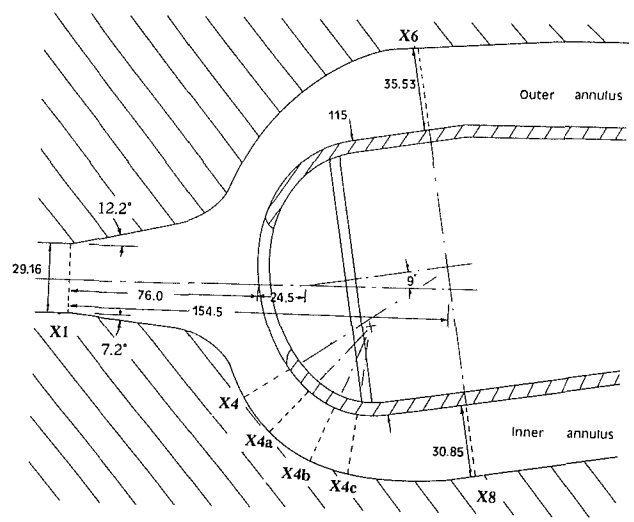


Fig. 2(a) Dump diffuser geometry



All dimensions in mm

Fig. 2(b) Faired diffuser geometry

a 14 percent reduction of overall system length. The performance of these systems is assessed in terms of the total pressure loss and static pressure recovery between prediffuser inlet and the annuli surrounding the flame tube head. Detailed measurements around the combustor head are also presented and used to identify the important processes and loss-generating mechanisms within both systems and thereby offer some explanation for any observed differences in performance.

Experimental Facility

Although of rectangular cross section, for ease of understanding the facility is described using terminology associated with the annular system it is trying to simulate. Hence, rectangular flow passages are described as annuli while at a given axial plane, locations are described in terms of radial and circumferential components. Furthermore, since the overall program objective was to investigate diffuser design concepts, all tests were performed without simulation of the burner feed arms, flame tube supports pins, and other detailed features that would be present within an operational system.

The test facility was manufactured predominantly out of Perspex, its modular design incorporating an inlet section, test section, and three exhaust pipes. Air was drawn from atmos-

Nomenclature

B = hot-wire calibration coefficient
C_p = static pressure recovery coefficient
E = mean hot-wire voltage
E_o = hot-wire voltage with no flow
h_o = prediffuser inlet height
h_1 = prediffuser exit height
K = button hook calibration factor
L = diffuser overall length
M = Mach number
n = hot-wire calibration coefficient
p = local mean static pressure
p_{ps} = local "pseudo" static pressure (Fig. 3)
P = local mean total pressure
q = dynamic head
R_n = nondimensional annulus height
r = radius relative to the center of curvature
t = static temperature
u_i = instantaneous velocity
u = local mean velocity
u' = fluctuating (rms) velocity component
α = kinetic energy flux coefficient
λ = total pressure loss coefficient
ρ = density

Superscripts

$\bar{\ }$ = mass-weighted mean value
$\tilde{\ }$ = momentum mix mean value

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phere into a plenum box (Fig. 1), located above the vertically mounted rig, prior to entering the inlet section and passing through the test rig. This air was then exhausted into a plenum box, below the facility, prior to being expelled to atmosphere. A centrifugal fan, driven by an 11 kW variable speed motor, was located in the exhaust system and provided the pressure rise required to draw air through the facility.

The scale of the working section was such that diffuser dimensions were typical of those likely to be found within a large turbofan engine. For the standard dump geometry a 2.5 deg outwardly canted prediffuser was incorporated of length 58 mm, inlet height 29.16 mm, and an area ratio of 1.45 (Fig. 2*a*). This was designed according to the current rules for diffusers [6] and projected into the dump cavity. The gap between prediffuser exit and the combustor (i.e., dump gap D/h_1) was approximately 1.0, with the flame tube head being canted outward by 9 deg. Head porosity was provided by holes of diameter 50 mm, the flow then passing through the cowl back plate prior to entering the combustor primary zone. The total width of the rig was 360 mm and consisted of four burner sectors, each sector containing a cowl hole. However, measurements were limited to the central sector, 90 mm wide, where the effects due to the presence of the sidewalls were at a minimum. The faired system (Fig. 2*b*) was 25.5 mm shorter, which was achieved by moving the flame tube closer to prediffuser inlet. Hence, the amount of overall diffusion between the inlet plane and the annuli surrounding the flame tube was the same for both designs. In addition, liners were attached to alter the profile of the inner and outer casings while also modifying the prediffuser geometry. It should also be noted that the diffuser systems incorporated slight variations in flame tube head geometry. In the dump system, sharp edged "lips" or shoulders on the combustor at entry to the outer and inner annuli reflected current engine hardware, their presence being due to flame tube manufacturing techniques. These were found to have virtually no effect on the performance of the dump system [5], and were not present for the tests on the faired system.

All measurements were obtained with low levels ($Tu < 1$ percent) of inlet turbulence and in the absence of compressor blade wakes. The distribution of mass flow to the flame tube, inner and outer annuli was controlled by throttles located in the downstream exhaust pipes. In addition to these three individually controllable flow paths, a further throttle in the exhaust system was used to control the total mass flow passing through the test rig. The facility was operated at a constant prediffuser inlet Mach number of 0.14, which corresponded to a Reynolds number based on the prediffuser inlet hydraulic diameter of 1.8×10^5. The mass flow split around the flame tube head was nominally set to

Inner annulus	43.1 percent
Flame tube	14.9 percent
Outer annulus	42.0 percent

This mass flow split was maintained within ± 0.3 percent of these values, thereby ensuring that the amount of flow diffusion to the outer and inner annuli was the same in all tests.

Instrumentation and Data Acquisition

The quantitative performance was mainly assessed using "button hook" pitot probes (Fig. 3), which could be inserted into the facility through holes of the same approximate diameter as that of the probe stem. Radial movement of each probe was controlled by a stepper motor, whereas a probe was circumferentially located either through discrete holes or small slots within which the probe and an associated blanking plate could slide. At any point in the flow the "button hook" pitot probes were used to obtain static as well as total pressures and this information was complemented by numerous wall static

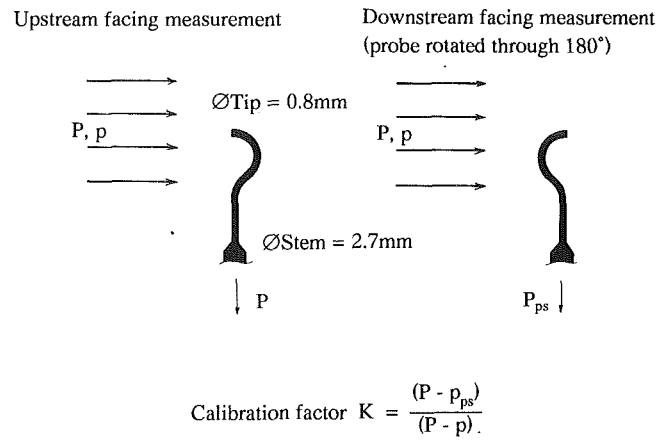

Fig. 3 Button hook probe operation

tappings located around the flame tube head and within the outer and inner annuli. In addition, constant temperature anemometry using a single hot-wire probe (Dantec type 55P11) was also used to give information on the turbulence levels within each diffuser system.

The overall performance of each system was assessed from measurements made at prediffuser inlet (X1) and the traverse planes farthest downstream within the outer and inner annuli (X6, X8). The axial location of these latter planes coincided with that of the primary port admission holes that would be present in engine hardware. At these planes some 20 data points were recorded for each radial traverse, this being repeated at 21 circumferential locations spaced over the 90 mm between burners. Measurements were also made at traverse planes X4 to X4c in order to indicate the nature of the flow passing around the flame tube head. However, in this case measurements were restricted to a single circumferential position located midway between burners, with approximately 20 data points being collected at each traverse plane.

All pressures were measured using transducers, which produced an analogue output signal proportional to the pressure measured. These voltage signals were digitized using an analogue to digital convertor (ADC), with the 5 second averaging of each signal being based on 250 samples at intervals of 20 milliseconds. Hot-wire voltage signals from a Dantec CTA 56C17 bridge circuit were similarly recorded, although these values were based on a 10 second averaging of the data.

Data Reduction

The curvature of flow around the flame tube head means that conditions in this region could not be defined using a pitot probe and wall static tapping. Instead, the technique outlined by Fry [7] was adopted in which the "button hook" probe, as well as being used as a conventional pitot probe, was also used to obtain local static pressure within the flow field. By recording the upstream (total) pressure and the value obtained when the "button hook" probe was rotated through 180 deg, into the downstream direction, the static pressure could be deduced with the aid of a suitable calibration. The calibration factor was assumed constant for a given probe and was defined as:

$$K = \frac{(P - p_{ps})}{(P - p)} \tag{1}$$

where P is the total pressure of the flow measured by the probe when facing upstream, p_{ps} is the "pseudo" static pressure recorded by the probe when pointing downstream, and p is the flow true static pressure (Fig. 3). Since the probe was calibrated prior to its use on the test rig, at a given point in

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the flow the true static pressure could be calculated from the measured values of P and p_{ps}. Calibration of each probe was performed in a low-turbulence calibration facility for which the conditions at the working section could be accurately defined. Over the speed range of 10 m/s to 40 m/s the values of K were within ± 1 percent of the mean value.

Calibration of each single hot-wire was based on the law:

$$E^2 - E_o^2 = Bu^n \tag{2}$$

where E is the measured voltage and E_o the "wind off" voltage of a hot wire with calibration coefficients B and n. Turbulence intensity was calculated using the formula:

$$Tu = \frac{u'}{u} = \frac{E_{\text{rms}}}{E^2 - E_o^2} \cdot \frac{2E}{n} \tag{3}$$

where E_{rms} is the root mean square of the fluctuating voltage component. The fluctuating (rms) velocity (u') was calculated from the turbulence intensity and the mean velocity at that point in the flow field.

At any measurement plane the mass flow rate and mean velocity were defined as:

$$m = \int \rho u dA = \rho \bar{u} A \tag{4}$$

while spatially averaged values of total and static pressure at each plane were derived using both mass-weighted and momentum mix techniques [8].

Mass Weighting

$$\bar{P} = \frac{1}{m} \int P dm = \frac{1}{\rho \bar{u}} \int P \rho u dA$$

$$\bar{p} = \frac{1}{m} \int p dm = \frac{1}{\rho \bar{u}} \int p \rho u dA \tag{5}$$

In addition, a kinetic energy flux coefficient (α) is defined that compares the kinetic energy flux of the measured velocity profile with that of a uniform profile with similar mass flow:

$$\alpha = \frac{\int \frac{1}{2} u^2 \rho u dA}{\frac{1}{2} \bar{u}^2 \rho \bar{u} A} \tag{6}$$

so that

$$\bar{P} = \bar{p} + \alpha \frac{1}{2} \rho \bar{u}^2 = \bar{p} + \bar{q}$$

It should be noted that it is the pressures derived from this technique that are used to define the performance of each diffuser system.

Momentum Mix. The spatially averaged total and static pressures obtained from this method are the values that would be recorded if a nonuniform flow mixed out to uniformity in a duct of constant area and in the absence of heat transfer and wall friction. Hence, the momentum mix total pressure is always lower than the mass-weighted value since it includes the pressure loss associated with the mixing out of the nonuniform profile. As outlined by Livesey and Hugh [8], the calculation of the equivalent uniform stream is based on continuity of mass, momentum, and energy flux.

Momentum:

$$\int p(1 + \gamma M^2) dA = \tilde{p} A (1 + \gamma \tilde{M}^2) \tag{7}$$

Mass:

$$\int \frac{p}{Rt} u dA = \tilde{p} \tilde{M} A \sqrt{\frac{\gamma}{R\tilde{t}}} \tag{8}$$

Energy Flux:

$$\int C_p \left(t + \frac{u^2}{2C_p} \right) \frac{p}{Rt} u dA = m C_p \tilde{t} \left(1 + \frac{\gamma - 1}{2} \tilde{M}^2 \right) \tag{9}$$

The three unknowns ($\tilde{p}$, $\tilde{M}$, $\tilde{t}$) can be obtained using these three equations, which allows the momentum mix total pressure to be calculated from compressible flow relationships. It should be noted that in this paper, these pressures are mainly used to assess the loss in the outer and inner annuli that is associated with the mixing out of the highly distorted profile in these regions.

Changes in the spatially averaged pressures between plane "b" and an upstream plane "a" within each diffuser system are expressed in terms of the total pressure loss (λ) and static pressure recovery (C_p) coefficients:

$$\lambda_{a-b} = \frac{P_a - P_b}{P_a - p_a} \qquad C_{p_{a-b}} = \frac{p_b - p_a}{P_a - p_a} \tag{10}$$

Note that these coefficients can be based on either mass-weighted or momentum mix spatially averaged pressures.

Information concerning the radial distribution of flow properties at a measurement plane is given in terms of pitch-averaged profiles, which are derived by averaging the relevant flow parameter in the circumferential direction. At a specific annulus height, a pitch-averaged velocity therefore represents the area-weighted mean value of the velocities measured at that height and is nondimensionalized by the overall mean velocity for that traverse plane. Around the flame tube head, the total pressure distributions are nondimensionalized with respect to the mass-weighted total pressure at that plane and the mass-weighted total and static pressures for the short faired design at plane X8.

Estimate of Experimental Errors

Mass-weighted pressures at the traverse plane X1, X6, and X8 were repeatable to better than ± 1 mm H_2O, a variation of less than ± 1 percent of the inlet dynamic head. This results in the derived total pressure loss and static pressure rise coefficients relating to overall diffuser performance being repeatable to better than ± 0.007 of a given value.

Because of flow curvature, turbulence levels, and the shape of the velocity profiles, it is extremely difficult to obtain an accurate definition of the flow parameters around the flame tube head. For example, the maximum velocities derived from the "button hook" probes at planes X4 to X4c were consistently greater, by up to 8 percent, than the values indicated by the hot-wire data. The relatively poor resolution of these data should therefore be considered when viewing the results. Based on these differences in velocity it has been estimated that the spatially averaged pressures in this region are only repeatable to within ± 4 mm H_2O, resulting in loss coefficients being accurate to within ± 0.03 of a given value. However, as will be subsequently shown using the measured pressure values obtained within each diffuser, it is thought the accuracy of these data is better than this estimate.

One problem associated with facilities that only simulate sectors of a fully annular flow field is that of the growth of boundary layers on each sidewall. Due to the diffusing nature of the flow field this is of particular significance in the present investigation and leads to the migration of fluid toward the center of the rig. Detailed measurements of these sidewall boundary layers indicated that at planes X8 and X6 up to 10 percent more flow was passing through the central burner sector, where diffuser performance was assessed, due to this effect. This reduces the amount of diffusion undertaken by the flow in the central measurement region, which will lead to errors in the absolute values of the performance parameters. However, measurements indicated the sidewall boundary layer effect was the same for each diffuser geometry tested therefore

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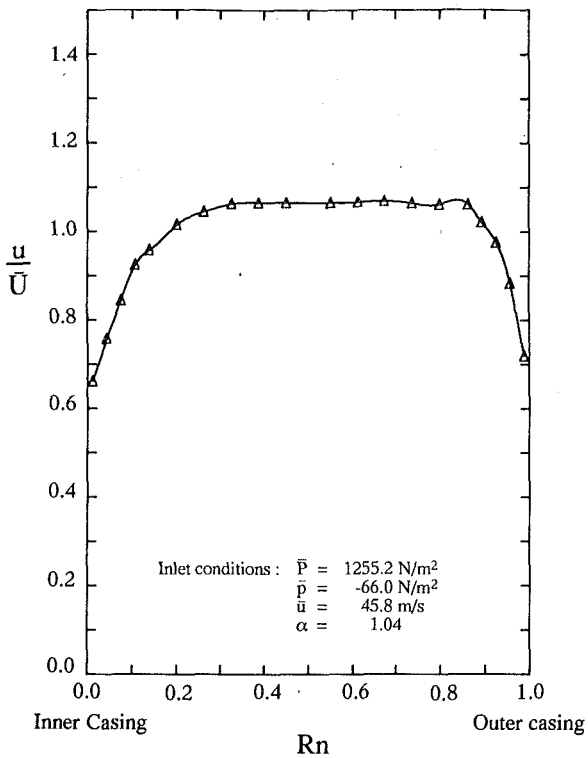

Fig. 4 Pitch-averaged velocity profile—diffuser inlet (X1)

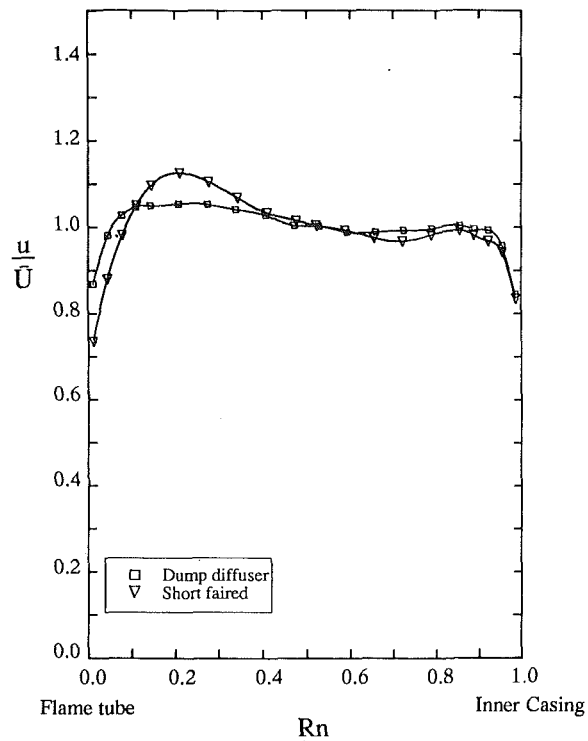

Fig. 5 Pitch-averaged velocity profiles—inner annulus (X8)

Table 1 Overall diffuser performance data (mass weighted)

	Dump diffuser	Faired diffuser
Outer annulus:	λ_{1-6} = 0.318	λ_{1-6} = 0.221(-30%)
	C_{p1-6} = 0.469	C_{p1-6} = 0.519
Inner annulus:	λ_{1-8} = 0.302	λ_{1-8} = 0.181(-40%)
	C_{p1-8} = 0.454	C_{p1-8} = 0.535
Flux coefficients:	α_1 = 1.04	α_1 = 1.04
	α_6 = 1.09	α_6 = 1.26
	α_8 = 1.02	α_8 = 1.04

allowing the relative performance of each system to be evaluated.

Results and Discussion

Inlet Conditions. Tests were performed with a clean upstream geometry resulting in relatively thin boundary layers (Fig. 4) at prediffuser inlet with no simulated compressor blade wakes. Turbulence intensity was less than 1 percent in the midpassage region. Stevens et al. [9] have already indicated the favorable mixing effects on diffuser boundary layers that can be produced when blade wakes are present. Within experimental error these inlet conditions were the same for the two geometries tested.

Overall System Performances. For the standard dump design, the mass-weighted total pressure loss (λ) to the outer (X6) and inner (X8) annuli was 0.318 and 0.302, respectively (Table 1). Allowing for differences in the geometry tested, these values are thought to be in reasonable agreement with previous results. For example, with no head porosity and a smaller flame tube depth ($W/h_o = 3.5$) Fishenden and Stevens [1] recorded loss values to each annulus of 0.24. In contrast, on a deeper flame tube ($W/h_o = 5.5$) but at a larger dump gap ($D/h_1 = 1.1$) Stevens et al. [31] recorded losses of 0.336 (outer) and 0.313 (inner). In these latter tests, however, the losses are thought to be relatively low, since performance was not evaluated from area traverses, but at a single circumferential location midway between burners.

Results from the faired diffuser show a significant reduction in total pressure loss and an associated improvement in static pressure recovery relative to the dump system (Table 1). For the flow passing to the outer annulus the mass-weighted pressure loss was reduced by 30 percent while a 40 percent loss reduction was observed for the inner annulus flow. This improvement in performance is, of course, in addition to the length reduction of 25.5 mm offered by the short faired design. For a dump diffuser this length reduction would have been obtained by an effective change in the dump gap (D/h_1) from 1.03 to 0.43. On tests with a flame tube of relatively small depth ($W/h_o = 3.5$), Fishenden and Stevens [1] observed a 15 percent loss increase for a dump gap reduction of this magnitude. However, Srinivasan et al. [4] using a flame tube of more comparable geometry ($W/h_o = 4.1$) indicated loss increases of between 30 and 60 percent depending on mass flow split.

The radial distribution of fluid within the outer and inner annuli of each diffuser system is shown by the pitch-averaged velocity profiles at the measurement planes (X6, X8). For both diffusers, the flow within the inner annulus appears to be relatively well mixed out (Fig. 5) with virtually no difference between the two systems. However, this is not the case within the outer annulus, where the profiles show a noticeable bias toward the flame tube (Fig. 6). Furthermore, the lower loss associated with the faired system has resulted in a more distorted profile in this region, and this is also confirmed by the kinetic energy flux coefficients (Table 1). It could be argued that this profile reflects an inherent loss that has yet to be realized, so that a relatively high pressure loss will occur within the outer annulus, downstream of plane X6, as the more distorted profile mixes out. However, estimates based on the momentum mix values indicate this could only result in the loss reduction for the faired diffuser falling from 30 to 27 percent.

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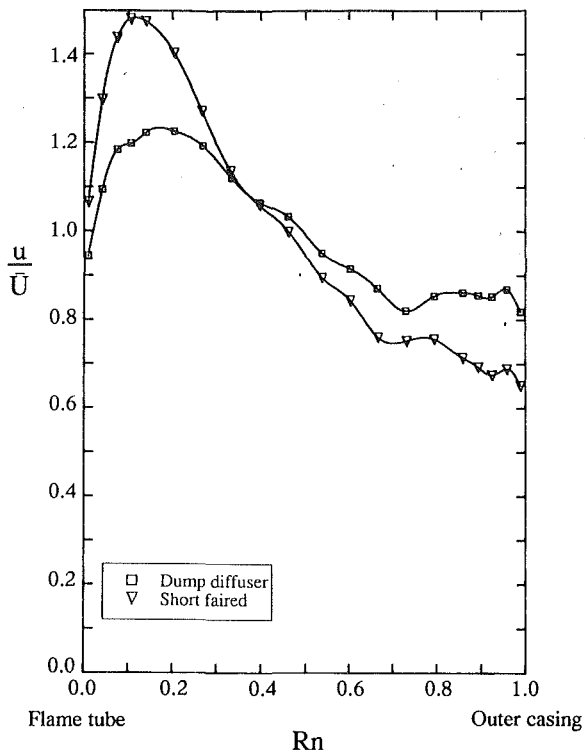

Fig. 6 Pitch-averaged velocity profiles—outer annulus (X6)

Table 2 Flame tube head pressure loss data

	Dump diffuser	Faired diffuser
Mass weighted :	λ_{1-2} = 0.03 (10%)	–
	λ_{1-4} = 0.09 (30%)	λ_{1-4} = 0.01 (5.%)
	λ_{1-4a} = 0.15 (50%)	λ_{1-4a} = 0.06 (30%)
	λ_{1-4b} = 0.21 (70%)	λ_{1-4b} = 0.10 (50%)
	λ_{1-4c} = 0.26 (87%)	λ_{1-4c} = 0.17 (85%)
	λ_{1-8} = 0.30 (100%)	λ_{1-8} = 0.20 (100%)
Momentum mix :	λ_{1-4} = 0.15 (48%)	λ_{1-4} = 0.05 (25%)
	λ_{1-4a} = 0.22 (71%)	λ_{1-4a} = 0.12 (60%)
	λ_{1-4b} = 0.26 (84%)	λ_{1-4b} = 0.16 (80%)
	λ_{1-4c} = 0.30 (97%)	λ_{1-4c} = 0.19 (95%)
	λ_{1-8} = 0.31 (100%)	λ_{1-8} = 0.20 (100%)

Detailed Diffuser Performances. Using a hot wire and button hook probe, attempts have been made to define the flow field around the flame tube head of each diffuser system and thereby account for differences in their overall performance. Measurements were restricted to the rig centerline, midway between cowl holes, and concentrated on the inner annulus flow, which exhibited the greatest differences in loss. Up to six planes were traversed within each system, although for reasons of brevity only a limited number of the measured distributions are presented here. It should also be noted that in the dump diffuser, no attempt was made to define the recirculating flow field within the dump cavity. Hence, for this system spatially averaged pressure values at each plane (Table 2) were only evaluated for the flow passing directly into the downstream annulus. The streamline dividing the mainstream and recirculating flow was defined as that which gave the required mass flow upon integration of the velocity profile between this location and the flame tube surface.

In the past, very few measurements have been made on the flow field around a flame tube head due to the difficulty of making accurate measurements in such regions. Concern has

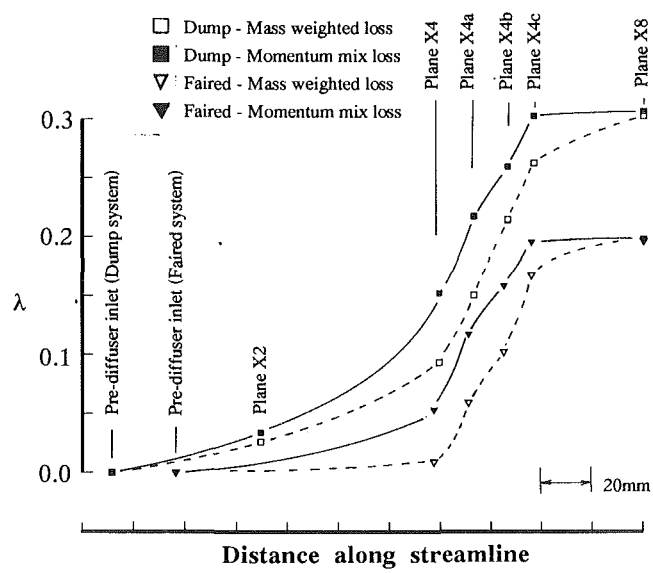

Fig. 7 Pressure loss (λ) through each diffuser system

also been expressed as to the quality of data obtained from this investigation, since differences in velocity of up to 8 percent were recorded between the hot wire and button hook probes. However, the mass-weighted and momentum mix pressure data for each plane (Table 2) appear to correlate reasonably well and agree with the measurements of relatively high accuracy at plane X8. This has been expressed graphically in terms of the position of each plane within the dump and faired systems (Fig. 7), their location being defined as the intersection point along a streamline passing down the center of the prediffuser and through the maximum velocity points around the flame tube head. It can be seen that both the mass-weighted and momentum mix pressure loss data lie on reasonably smooth curves with no significant discontinuities. This correlation of the data between planes therefore gives some degree of confidence in the pressure measurements and of the loss values quoted for each station.

Dump Diffuser. For the dump system approximately 10 percent of the mass-weighted total pressure loss ($\lambda_{1-2}=0.03$) occurs within the prediffuser and is confined to the fluid within the boundary layers adjacent to each casing. At prediffuser exit the flow then separates and rolls up to form a recirculation zone in the dump cavity. The size of this region is indicated by the velocity profile at plane X4 (Fig. 8*a*), with reattachment of the flow to the inner casing occurring in the region of plane X4c (Fig. 8*b*). This vortex in the dump cavity bounds the mainstream flow, which is discharged into the dump region and impinges on the flame tube head.

At plane X4 (Fig. 8*a*), the high-velocity region close to the flame tube surface indicates acceleration of the flow away from the stagnation point, with velocities in excess of 40 m/s being recorded. This acceleration continues around the flame tube head with the maximum velocity occurring in the vicinity of plane X4b. However, at greater distances from the surface, the velocity decreases rapidly, producing a highly sheared profile with kinetic energy flux coefficients in excess of 1.7. The distorted profile is mainly associated with the curvature of the flow that is required in order for it to pass around the flame tube head. Since flow curvature is still present at plane X4c the profile is still highly sheared (Fig. 8*b*), but downstream of this location the turning is complete with the flow entering the inner annulus. Comparison of the velocity distributions at planes X4c and X8 indicate a rapid mixing out of the profile in this region.

The measured fluctuating velocities were recorded with a

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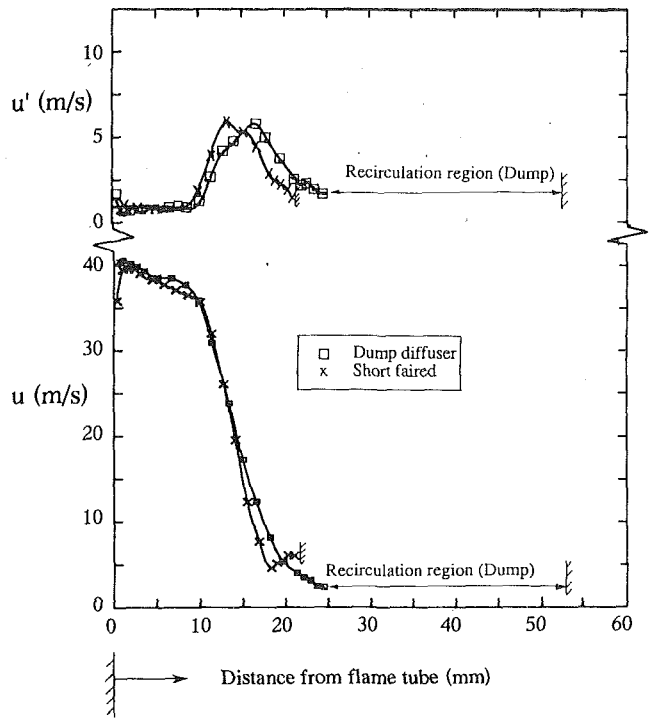

Fig. 8(*a*) Velocity profiles—plane X4

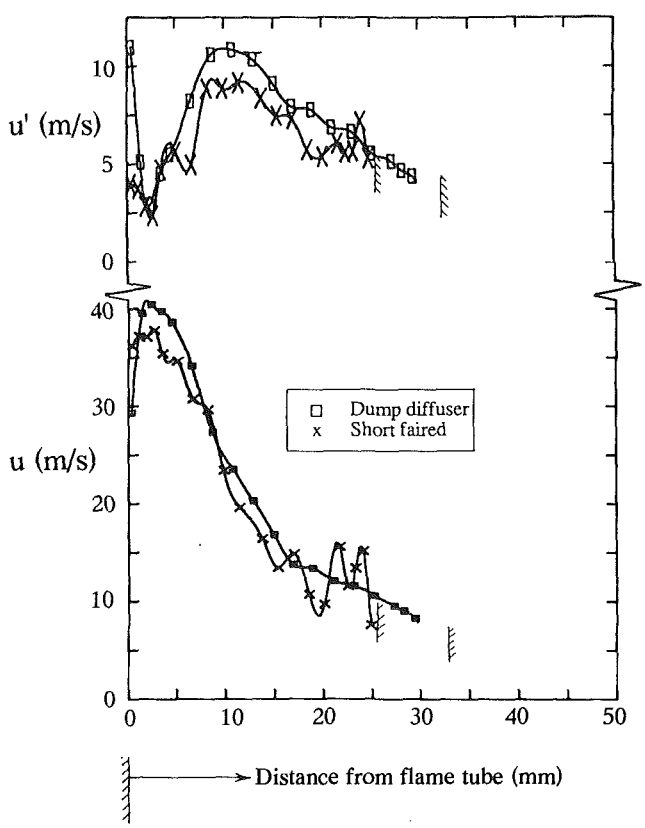

Fig. 8(*b*) Velocity profiles—plane X4c

hot wire, which was sensitive to the components both parallel and perpendicular to the flame tube head but not to the component in the circumferential direction. These velocities are therefore thought to be a good indication of the turbulent kinetic energy levels and at plane X4 (Fig. 8*a*) it can be seen that, as expected, this corresponds to the region of highly sheared flow. This continues to be the case around the flame

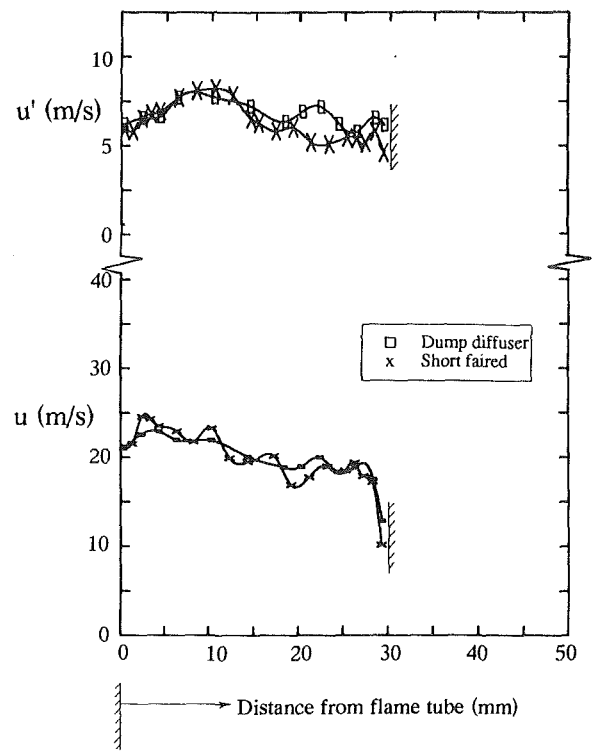

Fig. 8(*c*) Velocity profiles—plane X8

tube head, with the turbulent energy levels increasing steadily to reach a maximum value at plane X4c (Fig. 8*b*). In addition, the growth of the boundary layer on the flame tube head also produces turbulent flow, which can be observed at this plane. These high turbulence levels then assist in the rapid mixing out of the velocity profile between plane X4c and X8.

Since the turbulent energy is extracted from the mean flow there is a good correlation between regions of high turbulence and pressure loss. For example, approximately 30 percent of the system loss has occurred by plane X4, but the measured pressure levels close to the flame tube (1 mm–8 mm) are virtually the same as the maximum pressures recorded at prediffuser inlet (Fig. 9*a*). Hence, at this plane the loss is not associated with this high-velocity fluid but with the shear layer farther away from the surface. The fluid in this shear layer continues to exhibit large pressure losses farther around the head, with over 50 percent of the total system loss being generated within the relatively small region between plane X4 and X4c (Table 2). This region of sheared flow and relatively high pressure loss is thought to be associated with two important loss generating mechanisms:

(*i*) Dump loss: Energy has to be extracted from the mainstream flow passing through the diffuser in order to drive the recirculating flow in the dump cavity. This energy is extracted from the flow adjacent to the recirculation region, thereby increasing local velocity gradients and accounting for some of the observed turbulence and pressure loss.

(*ii*) Turning loss: As already observed, a heavily distorted velocity profile is observed as the flow is deflected around the flame tube head. This sheared flow must lead to the production of turbulent kinetic energy. However, as outlined by Fishenden and Stevens [1] this generation process will be significantly enhanced by the effects of streamline curvature. This is due to the unstable nature of a turbulent flow when the product of the local velocity and radius of curvature (ur) decreases with increasing radius of curvature (r). It can be seen that, apart from the boundary layer close to the flame tube head, the bulk

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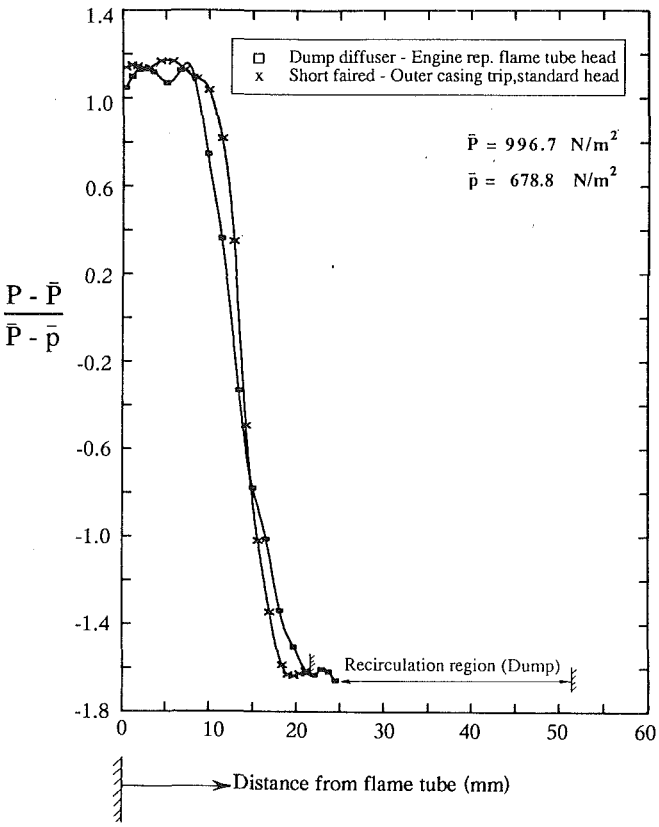

Fig. 9(*a*) Total pressure profiles—plane X4

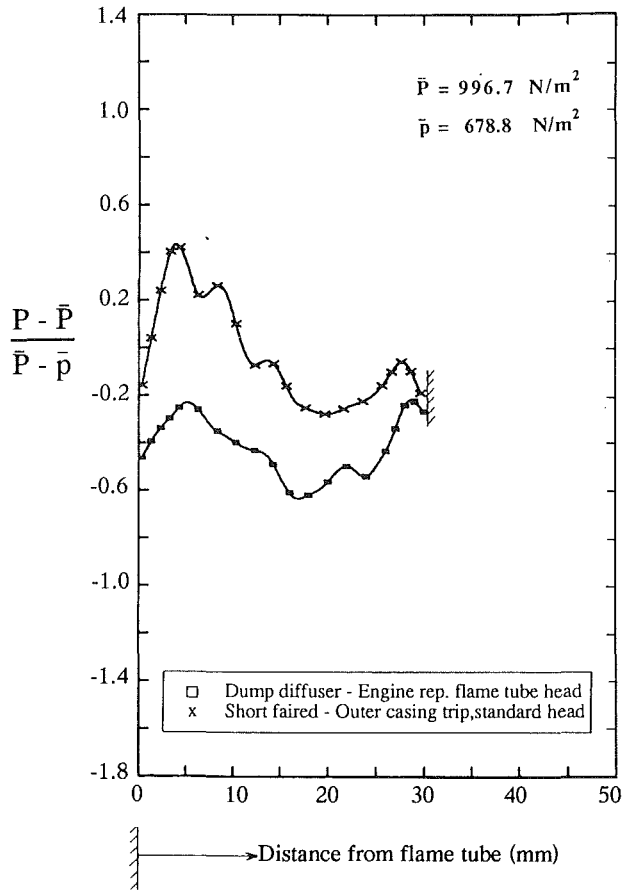

Fig. 9(*c*) Total pressure profiles—plane X8

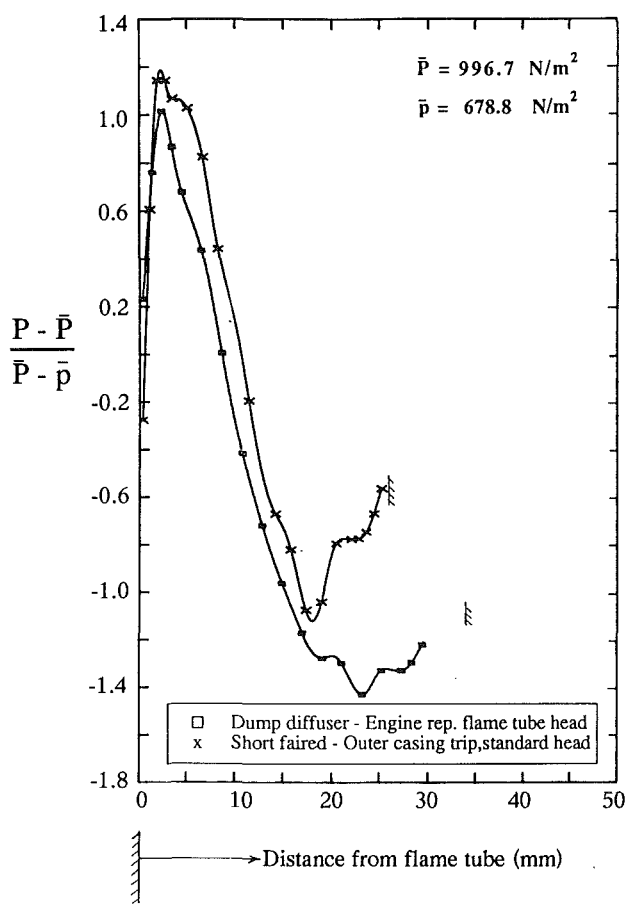

Fig. 9(*b*) Total pressure profiles—plane X4c

of the profile is unstable and will result in a significant enhancement of turbulent energy production.

Downstream of plane X4c a further loss in total pressure of approximately 10 to 15 percent occurs, with the large amounts of mixing in this region also producing a more uniform pressure distribution at X8. However, it can also be seen that between these planes virtually no change is observed in the levels of momentum mix total pressure (Table 2). Hence, the mass-weighted total pressure loss in this region is associated with the mixing out of the distorted profile that was observed at plane X4c and which was generated by the upstream turning of the flow.

Faired System. The faired system has no data available at plane X2 (prediffuser exit), but the total pressure loss up to plane X4 ($\lambda_{1-4}=0.01$) is significantly less than that observed for the dump system ($\lambda_{1-4}=0.09$). However, it should be remembered that within each diffuser system the geometry and flow conditions upstream of plane X4 are extremely different. The faired system incorporates a higher area ratio prediffuser, which may lower the dynamic head at prediffuser exit and thereby reduce losses around the combustor head. But also to be expected with this change in prediffuser area ratio is an increase in loss associated with this component. However, comparison of the dump prediffuser loss ($\lambda_{1-2}=0.03$) with the faired system measurement at plane X4 ($\lambda_{1-4}=0.01$) indicates the reverse effect. This is thought to be due to the flame tube influence on the upstream flow, which, as outlined by Stevens et al. [3], increases the static pressure at the center of a prediffuser passage while reducing the adverse pressure gradient along each casing. In this way prediffuser performance can be improved due to the lower maximum velocities in the passage center and the thinner boundary layers. For the faired system

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the closer proximity of the flame tube, due to its 25.5 mm movement toward the inlet plane, is therefore thought responsible for the reduction in prediffuser total pressure loss.

At exit from the prediffuser the close proximity of the flame tube along with the contoured casing profile avoids flow separation within the faired system. In fact, midway between cowl holes the flow initially undergoes a modest acceleration, which will further minimize pressure loss, prior to undergoing a controlled diffusion around the combustor head. This preturning of the flow may reduce local rates of curvature in this region, while the velocity profiles at plane X4 (Fig. 8*a*) show the elimination of the large dump recirculation within the faired system. A significant reduction in pressure loss will therefore occur since no energy is being extracted to drive the recirculating flow within the dump cavity. Comparison of the total pressure distributions at this plane (Fig. 9*a*) shows the reduction in mass-weighted pressure loss at plane X4 is associated with the flow between 9 mm and 15 mm from the flame tube surface. However, this difference in pressure loss is not reflected by the measured overall levels of fluctuating velocity and hence turbulent energy at this location (Fig. 8*a*). It is therefore thought that a significant proportion of the turbulent energy generated within the dump diffuser is transported into the recirculation region, which is not defined in the results presented here.

Approximately 80 percent of the total loss within the faired system occurs between planes X4 and X4c (Table 2), and it is in this region where flow turning and streamline curvature effects are dominant. However, the absolute loss value in this region ($\Delta\lambda = 0.16$) is virtually the same as that recorded for the dump diffuser ($\Delta\lambda = 0.17$). It has already been noted that reducing the dump diffuser length by 25.5 mm would significantly increase the system loss and this would be associated with higher velocities and steeper velocity gradients around the flame tube head. However, despite the shorter length of the faired system, at plane X4 the velocity profiles are remarkably similar to that of the longer dump diffuser (Fig. 8*a*). This continues to be the case farther downstream at plane X4c (Fig. 8*b*), where the fluctuating velocities are also reduced and the difference in pressure loss between the two systems is more apparent (Fig. 9*b*). The similarity of the velocity profiles around the flame tube head is also confirmed by the previous diagram (Fig. 7), which showed the mass-weighted and momentum mix pressures at each plane. The difference between these curves reflects the total pressure loss due to the distorted velocity profiles mixing out and is seen to be remarkably similar for each design. Thus, in addition to eliminating the energy loss required to maintain the recirculation within the dump cavity, it appears that preturning the flow at prediffuser exit and keeping it attached to the inner casing helps reduce the velocity magnitudes and gradients around the flame tube head. In this way, relative to the dump design, a length reduction of 25.5 mm has been achieved for the faired system without increasing the loss between planes X4 and X4c.

A further mass-weighted pressure loss ($\Delta\lambda = 0.03$) occurs between planes X4c and X8 (Table 2). However, as was also the case for the dump system, the momentum mix total pressures are virtually constant in this region, and so this loss is mainly associated with the mixing out of the distorted profile that was generated around the flame tube head. The fact that this profile appears not to be as well mixed out in the faired system is thought to be due to the lower upstream turbulence levels.

Stability of the Faired System. Although the dump diffuser system has always exhibited relatively high losses, its use has been widespread due to its inherent flow stability and insensitivity to manufacturing tolerances and thermal expansions. In contrast the faired system can be unstable, and this was illustrated in the present investigation by reducing the amount of air passing down the outer and inner annuli by increasing the flame tube mass flow. This off-design-point operation increased the amount of flow diffusion being attempted by the flow passing to each annulus and resulted in a flow separation that had no clearly defined location. This led to an instability with the flow switching between the inner and outer annuli and would have been an unacceptable occurrence under operational conditions [5]. In contrast the dump system has a clearly defined separation point at prediffuser exit and allows stable operation over a wide range of conditions. However, with the trend for flame tubes of greater depth and diffusers of shorter length so the difference in total pressure loss and static pressure recovery between the two systems will to continue to grow. It is thought the present results merely reflect this trend and so a point may be reached where the potential gains are sufficient to merit the use of a faired system so long as the instability problems can be addressed. Alternatively a hybrid design may be possible that offers some of the performance improvements of a faired system but that maintains the stability characteristics of a dump diffuser.

Conclusions

The performance of a standard dump diffuser system has been investigated and compared with a faired design that was 25.5 mm shorter. The following conclusions have been drawn:

- The mass-weighted total pressure loss coefficients (λ) between prediffuser inlet and the inner and outer annuli were reduced by 40 percent (inner) and 30 percent (outer) for the faired system. This improvement in overall performance is in addition to the 25.5 mm length reduction offered by this system.
- In the dump diffuser the flow separated at prediffuser exit and reattached to the inner casing at entry to the inner annulus. However, this recirculation loss was eliminated in the faired system with the flow remaining attached to the casing.
- Highly sheared velocity profiles were observed around the flame tube head in both systems due to the deflection and flow curvature in this region. These profiles were virtually the same within each system, despite the shorter length of the faired diffuser.
- The mass-weighted pressure loss within the inner annulus is associated with the mixing out of the profile generated around the flame tube head.
- Relative to the dump diffuser, the faired system is more prone to aerodynamic instabilities. Any reduction in pressure loss by using a faired geometry can therefore only be achieved after a full investigation of system stability.

This work is the subject of a continuing investigation involving both experimental measurements and computational predictions. The aim is to optimize diffuser geometries for a given length, combustor depth, head shape, etc.

Acknowledgments

This work was supported by Rolls Royce, Derby, Contract Nos. PVF1-250DM. The authors would also like to express their appreciation to Messrs. R. Marson, D. Glover, and L. Monk for their assistance in the numerous modifications incorporated into the test rig.

References

1 Fishenden, C. R., and Stevens, S. J., "Performance of Annular Combustor Dump Diffusers," *Journal of Aircraft*, Vol. 14, No. 1, Jan. 1977.

2 Young, K. F., Wray, A. P., and Stevens, S. J., "An Integrated Approach to the Design of Compressor OGVs and Combustor Pre-diffusers," Dept. of Transport Technology, Report No. TT89R03, Loughborough University, 1989.

3 Stevens, S. J., Wray, A. P., and Price, P. D., "The Aerodynamic Performance of a Modern Vaporising Combustor Dump Diffuser," Paper No. AIAA-88-3273, 1988.

4 Srinivasan, R., Freeman, G., and Grahmann, J., and Coleman, E., "Parametric Evaluation of the Aerodynamic Performance of an Annular Combustor-Diffuser System," Paper No. AIAA-90-2163, July 1990.

5 Carrotte, J., Denman, P., and Wray, A. P., "Performance Comparison

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of a Dump and Short Faired Diffuser System," Dept. of Transport Technology, Report No. TTR92R02, Loughborough University, 1992.

6 Carrotte, J., and Wray, A. P., "Research Into Short Combustor Diffusers," Dept. of Transport Technology, Report No. TTR91R03, Loughborough University, 1991.

7 Fry, P., Private Communication, Rolls Royce, Derby, United Kingdom.

8 Livesey, J. L., and Hugh, T., "Suitable Mean Values in One-Dimensional Gas Dynamics," *Journal of Mechanical Engineering Science*, Vol. 8, No. 4, 1966.

9 Stevens, S. J., and Wray, A. P., "The Influence of Blade Wakes on the Performance of Outwardly Curved Combustor Pre-diffusers," Paper No. AIAA-85-1291, 1985.

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